

HAMILTON RBG, ON, CanadaWMO: **712970**Lat: **43.2864N**Lon: **79.9052W**Elev: **335**StdP: **14.52**Time Zone: **-5.00 (NAE)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.3	5.3	-8.3	3.6	1.2	-2.8	4.8	7.2	16.8	27.2	15.1	25.8	4.1	250	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	89.3	73.4	86.2	72.2	83.3	70.4	76.4	85.4	74.5	82.7	72.7	80.1	5.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	127.3	81.3	71.9	119.3	79.3	70.2	112.3	77.2	40.1	85.6	38.2	82.8	36.6	80.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.3	11.0	-5.3	93.7	7.4	3.6	-10.6	96.3	-14.9	98.4	-19.0	100.4	-24.4	103.1		
(5)			-6.6	79.6	6.6	2.5	-11.4	81.4	-15.3	82.9	-19.0	84.3	-23.8	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	24.7	25.5	33.4	44.6	55.9	65.6	70.8	69.6	62.8	51.0	40.7	30.7
	DBStd	18.33	11.14	10.01	9.70	8.40	8.12	6.94	5.70	5.44	7.23	8.18	8.12	8.65
	HDD50	3210	787	685	520	206	28	0	0	0	2	89	294	599
	HDD65	6716	1250	1105	980	614	306	76	11	17	125	441	728	1063
	CDD50	2501	1	0	5	43	210	467	646	608	386	118	16	1
	CDD65	536	0	0	0	1	23	94	192	161	59	6	0	0
	CDH74	4734	0	0	0	28	264	872	1721	1304	504	41	0	0
	CDH80	1439	0	0	0	3	72	268	584	368	139	5	0	0
Wind	WSAvg	4.5	5.4	5.5	5.3	5.6	4.4	3.7	3.4	3.2	3.3	3.9	4.7	5.2
Precipitation	PrecAvg	34.8	2.8	2.4	2.5	3.0	3.1	3.1	3.2	3.0	3.0	2.9	2.9	2.7
	PrecMax	43.4	5.8	4.5	5.5	5.7	5.7	7.7	7.4	6.3	6.8	5.9	5.2	5.4
	PrecMin	27.0	1.2	1.3	1.1	1.2	1.0	0.9	0.7	1.2	1.0	0.7	0.8	1.4
	PrecStd	4.3	1.1	0.9	1.2	1.2	1.2	1.6	1.7	1.2	1.3	1.3	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	57.4	64.5	78.2	86.4	90.4	93.2	92.2	89.4	79.2	66.3	56.6
		MCWB	54.0	49.4	55.6	62.4	70.9	73.4	74.9	75.4	74.0	68.4	57.3	52.0
	2%	DB	48.5	48.3	58.4	70.9	81.2	87.1	89.8	87.0	84.0	73.2	60.9	50.6
		MCWB	45.9	44.9	50.7	58.1	66.2	72.4	73.5	72.9	71.5	64.2	54.7	47.5
	5%	DB	43.0	42.3	52.5	64.9	75.6	82.9	86.3	84.0	79.8	68.5	57.7	45.7
		MCWB	40.1	38.2	46.8	54.1	63.7	70.9	72.1	71.1	69.2	61.3	52.2	42.9
	10%	DB	38.8	38.4	46.9	58.9	71.0	79.0	82.9	81.0	75.8	64.1	53.5	42.1
		MCWB	36.7	35.3	42.2	50.8	61.0	67.3	70.1	69.6	67.3	58.4	49.0	39.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	49.2	57.8	64.9	72.5	77.2	79.6	78.5	75.6	71.2	59.4	52.9
		MCDB	56.2	53.5	62.2	74.1	82.1	86.5	88.7	85.5	85.4	77.1	62.6	55.8
	2%	WB	46.3	43.7	52.8	60.4	69.0	74.3	76.8	75.7	73.5	65.8	56.1	47.7
		MCDB	48.1	47.3	56.3	68.2	78.7	83.9	85.7	84.0	82.6	70.8	60.6	50.3
	5%	WB	40.4	38.3	47.5	55.8	65.5	72.1	74.4	73.6	71.1	62.2	52.5	43.3
		MCDB	42.5	41.1	52.7	63.7	73.1	80.6	82.1	80.8	77.0	67.3	56.6	45.3
	10%	WB	36.8	35.2	42.5	50.8	62.1	69.6	72.5	71.7	68.8	59.5	49.6	40.0
		MCDB	38.6	37.6	46.7	58.0	69.2	76.7	79.3	78.3	74.1	63.3	53.2	42.1
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.6	15.4	17.3	19.1	19.1	19.9	19.6	19.6	16.7	14.1	11.6
		MCDBR	15.7	18.9	22.6	28.0	25.8	24.1	23.4	22.8	24.5	22.6	20.3	15.5
		MCWBR	14.1	15.8	16.3	17.9	14.4	12.6	11.3	10.9	13.3	14.6	15.1	13.2
	5% WB	MCDBR	15.0	17.8	21.7	25.6	23.6	21.5	20.1	19.6	21.1	20.3	18.5	15.1
		MCWBR	14.2	15.6	17.1	18.7	15.0	12.4	11.3	10.8	12.6	14.1	14.8	13.7
Clear-Sky Solar Irradiance	taub		0.295	0.337	0.337	0.383	0.406	0.443	0.435	0.420	0.386	0.365	0.313	0.297
	taud		2.392	2.237	2.386	2.294	2.282	2.203	2.249	2.279	2.354	2.396	2.508	2.447
	Ebn at Noon		266	268	286	280	275	264	265	265	266	257	259	255
	Edh at Noon		24	34	33	40	41	45	43	40	34	29	22	22
All-Sky Solar Radiation	RadAvg		447	687	1082	1407	1714	1879	1877	1649	1318	796	496	340
	RadStd		45	72	116	154	174	131	134	85	123	74	66	40